

| EVOLVED SKILLS

*A design document for **oneblockmc**.*

| OVERVIEW

This document covers a custom core for a one-block gamemode, which needs to be built to work with the open-source plugin **SuperiorSkyblock**.

The main components of this system are as follows:

- **"Milestones and Skill Level"** feature where there are 3 skills, each with a set of "Milestones", tiered objectives each tied to the harvest of a specific resource type. Completion of each tier of these milestones awards "Skill XP" for the skill they're tied to, and once the player has earned enough skill XP, the skill will level up, unlocking/expanding access to skill perks and perk tiers.
- **"Skill Perks"** feature where all the other aspects of this system (one block generators, resource nodes, spawners, etc.) have tiered upgrades purchasable with custom items and money.
- **"One Block"** feature where each player island generates with a regenerating block. When a "one block" is broken, it regenerates again instantly as a new block. It will support dropping custom items, and be impacted by perk upgrades determining things like drop chances.
- **"Resource Nodes"** feature where players can craft placeable resource nodes that when broken give a specific resource, like iron nodes that drop raw iron each time they're mined. Once broken, a node regenerates after a configurable amount of time has passed.
- **"Upgradeable Spawners"** feature where players can craft placeable mob spawners that summon animals/monsters and are impacted by upgrades from the custom upgrade system that ties into all parts of this evolved skills system.

- **“Daily Growth Limits”** feature where each type of crop and sapling has a unique and expandable limit that once reached will prevent it from increasing in growth stage that day.
- **“Resource Generators”** feature that allows the player to craft placeable generators that produce resources over time, and can be upgraded. They passively produce resources and items while the player is online and on the island.
- **“Daily Challenges”** feature where the same tracking/objectives from milestones are utilized to give the player a set of daily objectives they can complete for rewards.
- **“Daily Bonuses”** feature where a more basic set of things are tracked (ores mined, mobs killed, crops harvested) and rewards are given at increasing thresholds that reset daily.
- **“Player Level”** feature where players earn player XP from milestones, daily challenges, and daily bonuses to increase their player level. Each level unlocks claimable rewards, and once the player reaches level 100, they can prestige at the cost of ingame money to reset their player level to 0, and earn all the rewards again.
- **“Player Journey”** feature acting as a long-term tutorial guiding the player through the progression in steps, tracking the objectives from milestones + some additional more generic ones. Completion of each step earns the player claimable rewards.

ONE BLOCK CORE

Each player island generates with a “one block”, a special, infinitely regenerating block where each time you break it, another block takes its place. Sometimes, the block is generated with another block on top of it (such as farmland with a crop on it), or an entity on top of it (like a cow or a zombie).

What blocks can generate are based on the player’s **Skill Perks** (covered below) which will unlock and allow new types of blocks to appear. Blocks are all set up in a config file, as entries. The following information should be able to be specified per entry:

- Block type (what physical block it is, such as "dirt" "iron_ore", etc.)
- Bonus type (optionally added to an entry, should allow the specification of a second ID for a block like "sugar_cane" "rose", etc.)
- Entity type (optionally added to an entry, should allow the specification of an entity ID like "zombie", "cow", etc.)
- Entry weight (the chance for this entry to be selected as the next block, assuming the entry requirement has been met. Uses a weight system for odds)
- Entry requirement (the perk tier of the "skill access" perk the player must have in order for this entry to be possible to appear, like "farming 8"

I SKILL LEVEL AND MILESTONES

Each player has **3 Skills; Mining, Farming, and Slaying**. Each has **50 Levels**, and are progressed with the **Skill XP** players earn from reaching new **Milestone Tiers**.

Each skill has a set of **Milestones**, each with **20 Tiers**. **Milestones** are progressed by harvesting their corresponding resource.

- For **Mining**, this includes mining ores, stones, and other blocks.
- For **Farming**, this includes harvesting logs, crops, and certain plants.
- For **Slaying**, this includes killing animals and monsters.

It's important that progress is only earned in the following circumstances:

- For **Ores, Stones**, and other blocks that come from **Resource Nodes**, only when the player harvests a **Resource Node** of that type, or the **One Block** itself.
- For **Crops**, only when the player harvests a **grown** crop, not a crop placed by the player (relevant for things like pumpkins, melons, sugar cane, etc. Anything that can be placed in a fully grown state should be considered and prevented.)

- For **Logs**, only when the player harvests a **grown** tree, not a log placed by the player.



When the player reaches a new **Milestone Tier** by meeting its **Requirement**, they should be sent a message in chat informing them, that when clicked, opens up the corresponding **Milestone Menu**.

You've reached Tier 3 for your {Type} Milestone! (Click here to view)

Claiming a **Milestone Tier** grants the player **Skill XP** that goes towards their **Skill Level**, and **Player XP** that goes towards their **Player Level**. If the player makes additional progress towards a milestone before claiming it, it should still be tracked as overflow, then added to the next tier once claimed.



The following **Milestones** exist, shown in the menus below:

- When the player has not yet unlocked a resource (based on their **Resource Access** perk) it should have a gray dye icon instead, or maybe a custom "locked" icon if one gets made





For each skill, there should be a config file where milestones can be added. Each milestone should be able to have the following information configured about it for each of its 20 tiers:

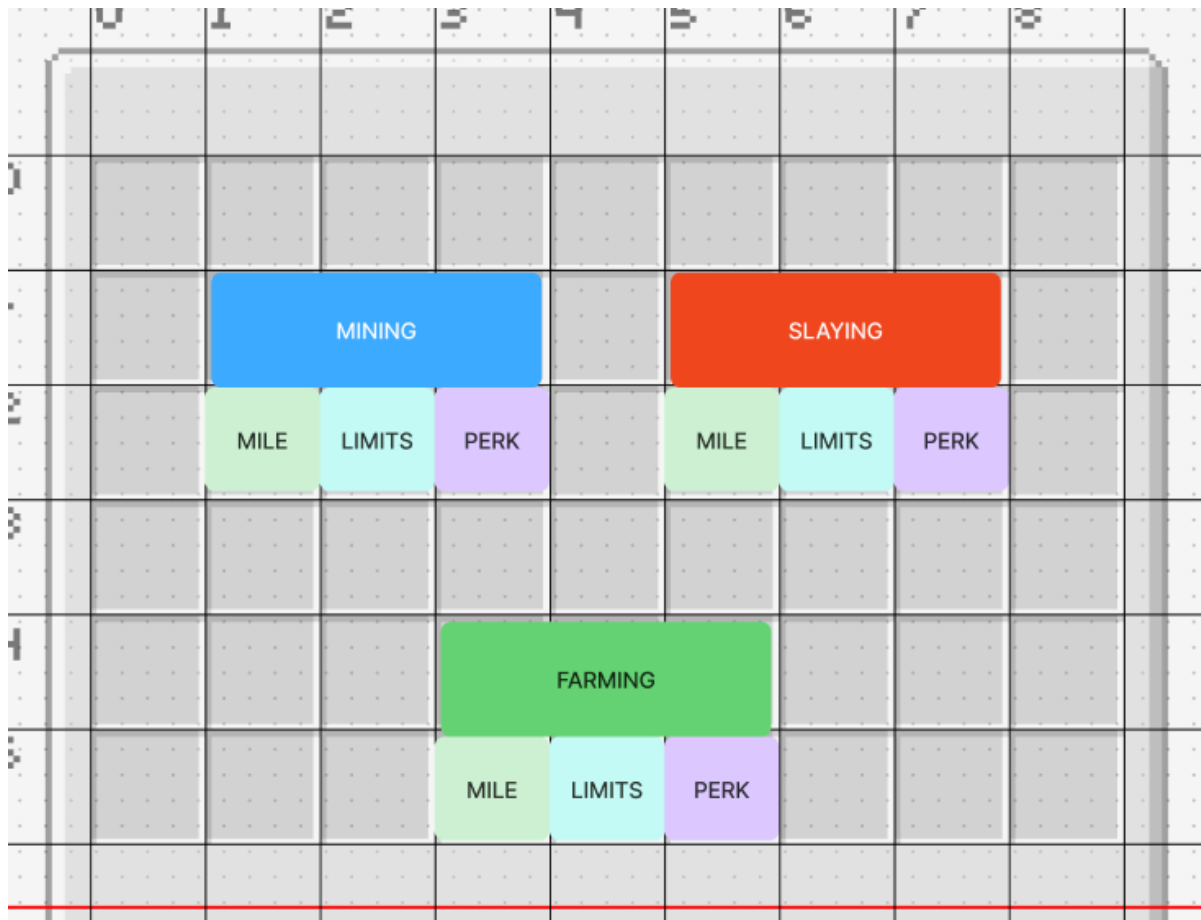
- **Tier Requirement** - How many times the milestone needs to be progressed to reach that tier.
- **Tier Skill XP** - How much skill XP is earned for reaching that tier.
- **Tier Player XP** - How much player XP is earned for reaching that tier.

Skills should also each have a config file where the skill level itself can be configured. For each of the 50 levels, the following information should be configurable:

- **Skill XP Requirement** - How much XP it takes to reach that tier.
- **Perk Unlock** - Which perk tier unlocks for reaching this skill level.
(More on perks next)

All menus related to **Skills** will be accessible in a single sub-menu. This will be a menu with a custom art background and custom buttons, the format can be seen in the image below. The 1x3 "Mining" "Slaying" and

“Farming” bits are headers, not buttons. The 3 buttons for each of the 3 menus per skill can be found and accessed below those headers.



SKILL PERKS

Each skill has a set of **5 Perks**, each with **10 Levels**. 4 of the perks function identically for each skill, and the final one ties into the unique aspect of the skill (resource nodes, spawners, or growth limits).

The perks for each skill are as follows:

- **Resource Access** - Which resources from that skill the **One Block** can spawn. See the above section on the **One Block** and the **Entry Requirements** for more context on this.
- **Resource Fragment Chance** - The chance for resources from that skill to drop its **Fragment** when harvested. **Fragments** are the

items used to craft nodes/spawners and expand the daily growth limits for crops, as well as expand the placement limits for nodes/spawners (more on this below).

- **Treasure Chance** - A multiplier on the chance for the player to get **Treasure**, a configurable bonus item, from harvesting a resource of that skill. The base chance is based on the resource, but this perk multiplies that chance. More on treasure below.
- **Generator Limit** - The number of generators for that skill the player can place on their island. More info on generators below.

The unique perks for each skill are as follows:

- **Mining Resource Node Speed** - Decreases the amount of time it takes for **Resource Nodes** to regenerate by a specified percentage, for example 40% would mean the **Respawn Timer** for all **Resource Nodes** would be 60% of its normal amount.
- **Farming Crop Growth Speed** - Increases the speed at which **Crops** and **Saplings** grow by a percentage. For example 10% would mean crops grow at 110% speed, as in 10% faster than usual.
- **Slaying Spawner Speed** - Increases the speed at which spawners summon mobs. For example 10% would mean crops grow at 110% speed, as in 10% faster than usual.

All 15 of these perks can be found in the **Skill Perks** menu, and each has **upgrade requirements** based on skill level.

- When configuring **Skill Level**, you can specify with perk at which tier unlocks when each level is reached.
- When the player has reached the necessary **Skill Level**, they can pay the **Upgrade Cost** (a configurable amount of money) required to upgrade that perk to the next tier.

This means that for each perk, the following things can be configured:

- Upgrade cost (in money) for each tier.

- Effect for each tier. (Like the percentage increase per tier for a speed increase, for example)



RESOURCE NODES

Resource Nodes are special blocks the player can craft and place. When placed, they become bedrock, until their **Respawn Timer** ends and they become a certain **Block**. For example, an **Iron Resource Node** would become an **Iron Ore**, that when broken would act just like an **Iron Ore** generated from the **One Block**. When harvested, it goes back to bedrock until its **Respawn Timer** ends again.

- Different **Resource Nodes** have different **Respawn Timers**, and these timers only count down while the player is online and loading them in.
- If the player is **Shifting** while mining the node, it will **Break** it instead of harvesting it, giving them the node back so they can move it and place it elsewhere.

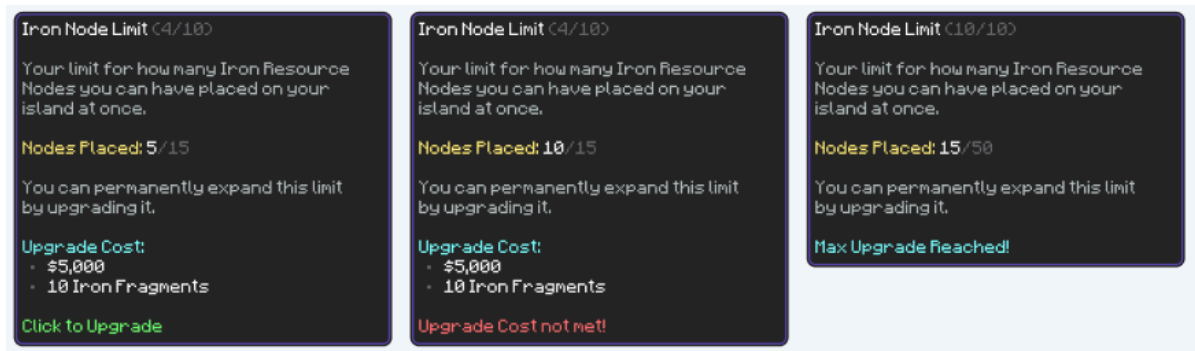
Each type of **Resource Node** (e.g. "Iron", "Gold", "Andesite", etc.) has a **per-island-limit** for how many can be placed.

- When this limit is reached (for example the player has a limit of **10 Iron Resource Nodes**, and they have 10 placed) and the player tries to place a **Resource Node**, it will not allow them to, and send them a message in chat informing them why, and clicking it will open the menu.

- You've reached the limit of {type of node}. Expand this limit in your **Mining Limits** menu ([Click to view](#))

These limits can be expanded by the player spending money and **Resource Fragments** of the corresponding type to permanently raise them.

- These limit increases have tiers/stages, where each tier increases the limit by X amount and costs Y amount of money and resource fragments.
- They need to be configurable per type of node.
- The menus for increasing these limits are identical in design to the milestone ones, just with different tooltips and interactions.



UPGRADEABLE SPAWNERS

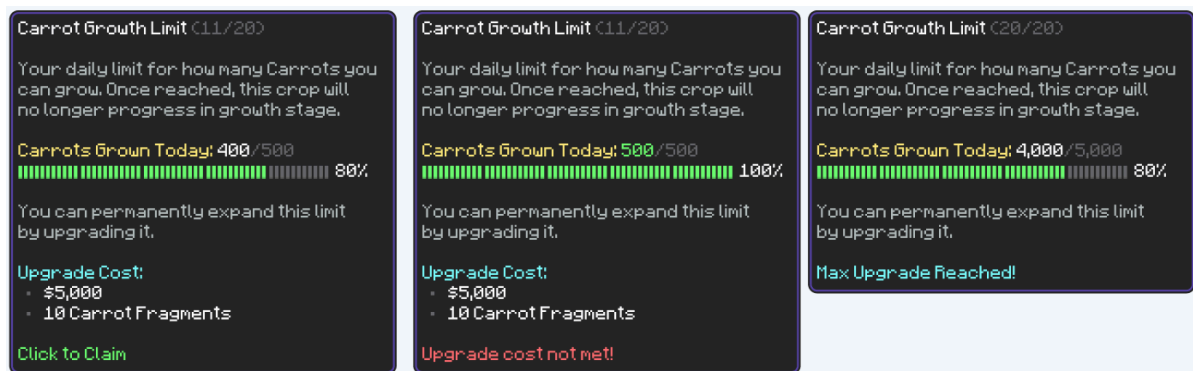
These spawners function almost identically to vanilla, but have a few integrations into the systems here that makes them need to be somewhat custom.

- As mentioned above, the **Slaying Spawner Speed** perk increases the speed of all spawners by a percentage.
- There are also **per-island-limits** identically to the system for **Resource Nodes** that players can expand through spending **Resource Fragments** of that spawner type.
- The tooltips and menus for this look identical to the ones for resource nodes.

DAILY GROWTH LIMITS

Daily Growth Limits are per-crop and per-sapling limits on how many of that crop/sapling can be grown that day. Once the limit is reached, that crop/sapling will no longer receive growth stage updates, and will remain frozen in time until the next day occurs, or the limit is expanded.

- Similar to expanding the limits for **Resource Nodes** or **Spawners**, there is a menu where players can expand these limits for each of the crops at the cost of money and resource fragments.
- The config here is identical to the one for those two systems, where I can configure the **Limits** and **Costs** for each “tier” of a resource's limits.



RESOURCE GENERATORS

Resource Generators are blocks that passively produce resources that can be claimed within their menus. Each skill has a different type of generator that produces a different type of resource.

- The player has a limit per type of generator (so per skill) on how many generators they can have placed. (Such as “10 Farming Genetators”). This limit is based on the corresponding skills **Generator Limit** perk.
- Generators produce resources at a set rate of 1 per minute. What resource is produced is randomly chosen, based on the player's current **Resource Access** entry weights, but only ones tied to that skill. (see the above one block section). Essentially, the generator is

like a passive version of the player's one-block, that produces resources specific to a skill.

- Generators store their produced resources in a chest inventory. When the chest is full, the generator stops producing.
- Generators look like a block with a hologram above them. The hologram simply says "Mining Generator" or farming or slaying. Clicking on the block is what opens the menu.

| TREASURE

Treasure is a generic term for **configurable bonus items** that can drop when harvesting the **One Block**, **Resource Nodes**, **Spawner Mobs**, **Crops**, or **Trees** from saplings.

I need a config file where I can specify which resources (mobs/ores/etc.) are hooked up to what **Treasure Loot Pools** and what the chance to pull from each pool is. This chance is run each time the player harvests a resource that has at least one pool tied to it.

The contents of **Treasure Loot Pools** are configured ingame, by adding items into a chest (as an admin), looking at the chest, and running the `"/treasurepool save (name)"` command, which registers it as a pool. Running the command with the `"(name)"` will override the previously saved pool.

You can also do `"/treasurepool view"` to see a menu containing all saved treasure pools, as chests with names. Clicking on a chest in this menu will open up a menu showing a view of the chest as it was saved. You can add and remove items in this view.

| PLAYER JOURNEY

The **Player Journey** is a tutorial system that is meant to guide the player through the server's progression.

It has the player complete **Tasks** in an ordered list, where they must complete the previous task in order to progress to the next one.

The tasks have **Requirements** (what the player must do) and **Rewards** (what they earn when they claim the completed task)

As shown in the menu example below, the icon for a task is:

- Green when completed
- Yellow when in progress
- Red when locked

Tasks are configured in a config file, where the following information can be specified per task. Tasks appear in the menu in the order they are configured in the config.

- **Task Requirement** - What the player needs to do to complete the task. This may need to support other actions in the future, but for the moment it consists of
 - Run a command - Specify a command the player needs to run
 - Craft an item - Specify an item the player must craft
 - Harvest a resource - Using the same tracking and options as milestones. (Such as "zombie", "iron_ore", etc.)
 - Reach X Milestone Tier - Checks if the player has reached X tier of Y milestone (Such as "zombie 20")
 - Reach X Skill Level - Checks if the player has reached X skill level
 - Complete Daily Challenge - Progresses if they player has completed a daily challenge (info below)
- **Task Amount** - How many times the player needs to complete the requirement in order to complete the task.
- **Task Reward** - What the player earns upon the completion of the task. This will always be specified as money or resource fragments from this system. Some tasks may also give **Player XP**, an asset that progresses the **Player Level** covered below.

Journey Task #11

Mine Iron Ore on your island.

Task Progress: 50/50

Rewards:

- \$5,000
- 10 Iron Fragments

[Click to Claim](#)

Frame 3159

Journey Task #11

Mine Iron Ore on your island.

Task Progress: 50/50

Rewards:

- \$5,000
- 10 Iron Fragments

Task claimed and completed.

Journey Task #11

Mine Iron Ore on your island.

Task Progress: 15/50

Rewards:

- \$5,000
- 10 Iron Fragments

Requirements not met!

Frame 3158

Journey Task #11

Mine Iron Ore on your island.

Task Progress: 0/50

Rewards:

- \$5,000
- 10 Iron Fragments

Task Locked!

Player Journey

Inventory

PLAYER LEVELING

The player's **Player Level** is progressed from the **Player XP** they earn from claiming **Milestones**, **Journey Tasks**, **Daily Challenges**, and **Bonus Tasks**.

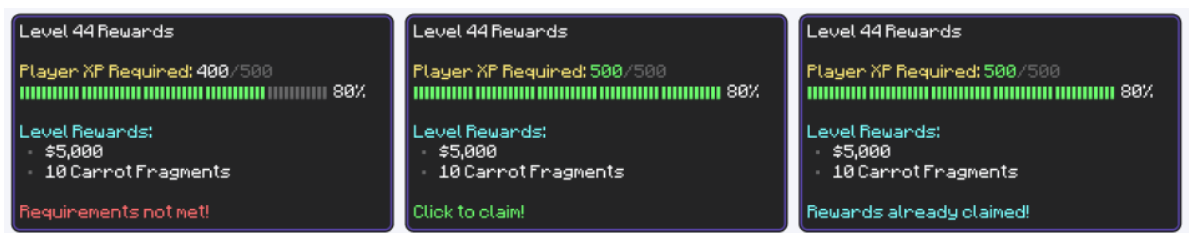
This level is shown next to their name in chat. Player level goes from 1-100, and for each level they reach, they can claim a set of rewards from the **Player Level** menu. Rewards show as an empty minecart when claimed, a chest minecart when claimable, and a tnt minecart when locked.

Once players reach level 100, they can also **Prestige**, resetting them to level 1. This costs a configurable amount of **Money**.

All earned **Player XP** can be **Multiplied** by a multiplier based on the player's rank. Each multiplier (1.2x, 1.3x, 1.5x, 2x, 2.5x, 3x) should be a permission node we can assign to a rank.

Player Level is configured in a config file, where the following info can be specified per level.

- **XP Requirement** - How much **Player XP** the player must earn to reach this level
- **Level Rewards** - What the player earns from claiming this level. This will always be specified as money or resource fragments from this system.





DAILY CHALLENGES & TASKS

Players have a **Daily Tasks** menu that consists of 3 core **Daily Tasks** and up to 10 **Daily Challenges**.

The three daily tasks are as follows:

- Kill Mobs (Any animal/mob from a spawner or the one block counts towards this)
- Mine Resources (Any block from a resource node, grown tree, or the one block counts towards this)
- Harvest Crops (Any crop from the resource node, or grown on the island counts towards this)

What needs to be configurable per task is its **stages**, which determine the rewards and requirements. Each task has **10 stages**, and each should be configurable in a config file.

- **Task Requirement** - How many times the player needs to complete the action at that stage in order to complete it.

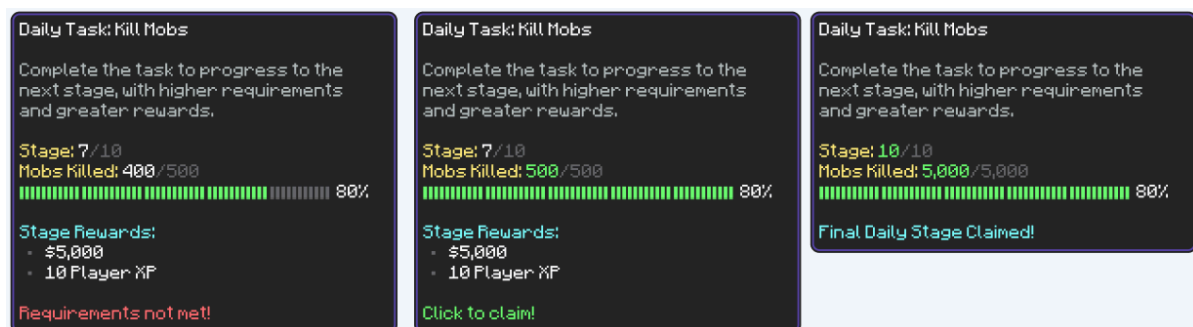
- **Task Rewards** - What the player earns from claiming this stage when completed. This will always be specified as an amount of **Player XP**, and money or resource fragments from this system.

This menu also has **Daily Challenges**, which are identical to **Daily Tasks** in function but do not have stages, and have more specific and randomized requirements.

The number of **Daily Challenges** the player has access to is based on their **Player Rank**. This can be handled with **permissions**, where each **Daily Challenge Slot** has a permission string that can be attached to a rank by a separate system.

What **tasks** are possible is based on the config, where any task can be added, with the following information. Their tooltips are identical to daily tasks, just without the "stage:" bit.

- **Task Requirement** - What the player needs to do to complete the task. Using the same tracking and options as milestones. (Such as "zombie", "iron_ore", etc.)
- **Task Amount** - How many times the player needs to progress the requirement in order to complete the task.
- **Task Reward** - What the player earns upon the completion of the task. This will always be specified as money or resource fragments from this system. Some tasks may also give **Player XP**.



Daily Tasks

Inventory